

Practical Network Defense

Master's degree in Cybersecurity 2025-26

Forward proxy activity

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Squid activity: as a forward proxy



To do the activities

- We will use Kathará (formerly known as netkit)
 - A container-based framework for experimenting computer networking: <http://www.kathara.org/>
- A virtual machine is made ready for you
 - https://drive.google.com/file/d/1W6JQzWVyH5_LKLD20R6XH1ugPDP5LWP5/view?usp=sharing
- For not-Cybersecurity students, please have a look at the Network Infrastructure Lab material
 - http://stud.netgroup.uniroma2.it/~marcos/network_infrastructures/current/cyber/
 - Instructions are for netkit, we will use kathara



The kathara VM

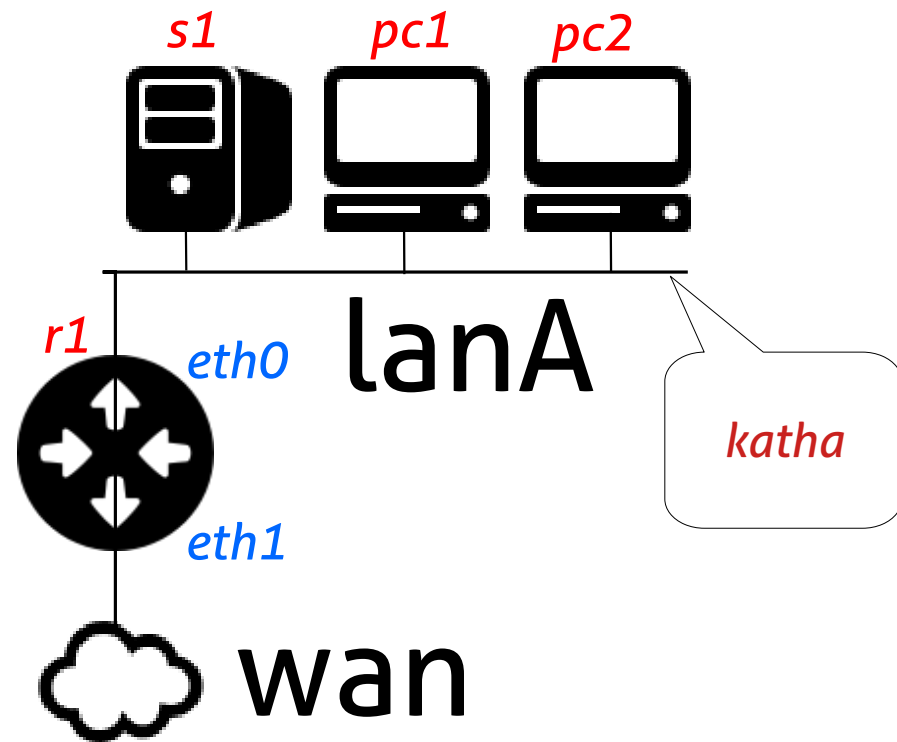
- It should work in both Virtualbox and VMware
- It should work in Linux, Windows and MacOS
- There are some alias (shortcuts) prepared for you
 - Check with **alias**
- All the exercises can be found in the git repository:
 - <https://github.com/vitome/pnd-labs.git>
- You can move in the directory and run lstart
 - **NOTE:** launch docker first or the first lstart attempt can (...will...) fail



Lab activity: lab6/ex1

pnd-labs/lab6/ex1: squid proxy

- In this lab you have to incrementally build the squid configuration
 - You can join lanA using the `connect_lab` script
- Refer to this [resource](#)
 - Most of the activity can be solved looking at the above resource
- Firstly, in r1 enforce the policy that only the proxy can use http and https (and obviously DNS) with iptables
 - Verify that pc1 and pc2 cannot use internet
- Take a look at the simple squid configuration file at `/s1/etc/squid/squid.conf`





Activity 1

- Configure pc1 and pc2 to use the squid proxy
 - `export http_proxy=192.168.100.80:3128`
- To start (resp. stop) squid: `systemctl start squid`
- Verify you can connect with http to a website (that uses http!)
 - Ex: <http://httpforever.com/>, <http://neverssl.com>, <http://http.badssl.com>, <http://detectportal.firefox.com/success.txt>, http://google.com/generate_204
 - Check with wireshark what happens
- Modify the squid.conf so that pc2 can not use http
 - Check with wireshark what happens
- Modify again the squid.conf to use a file with blacklisted websites



Activity 2

- Configure squid so that it can also allow https
- pc1\$ export https_proxy=192.168.100.80:3128
- To work, this requires the use of the CONNECT method
 - Reference https://wiki.squid-cache.org/SquidFaq/SecurityPitfalls#The_Safe_Ports_and_SSL_Ports_ACL
 - Extra details are provided in the original squid.conf file, found at s1/etc/squid/squid.conf.bak
 - Look for “SSL_Ports” and “Safe_ports”
 - When done, check with wireshark what happens



Activity 3

- Configure squid so that it requires the users to authenticate with username and password
- You can find more info about authentication methods on this resource:
 - <http://www.squid-cache.org/Doc/man/>
- You can use the ncsa method



Activity 4

- Configure squid to perform SSL Bump, in order to intercept the https traffic generated by the client pc1
- Reference:
 - <https://wiki.squid-cache.org/Features/HTTPS>
 - <https://wiki.squid-cache.org/ConfigExamples/Intercept/SslBumpExplicit>



Activity 5

- Configure squid and the topology to realize the configuration of a transparent proxy



That's all for today

- **Questions?**
- See you next lecture!
- References:
 - Ari Luotonen, Kevin Altis, [World-Wide Web Proxies](#), 1994
 - http://httpd.apache.org/docs/current/mod/mod_proxy.html
 - https://en.wikipedia.org/wiki/Proxy_server
 - Classical vs Transparent IP Proxies, [RFC 1919](#)
 - SOCKS 5, [RFC 1928](#)
 - HTTP 1.1, [RFC 7230](#)
 - [Policy based routing](#) and [Linux advanced routing and traffic control](#)
 - ICAP, [RFC 3507](#)
 - <https://wiki.squid-cache.org/FrontPage>